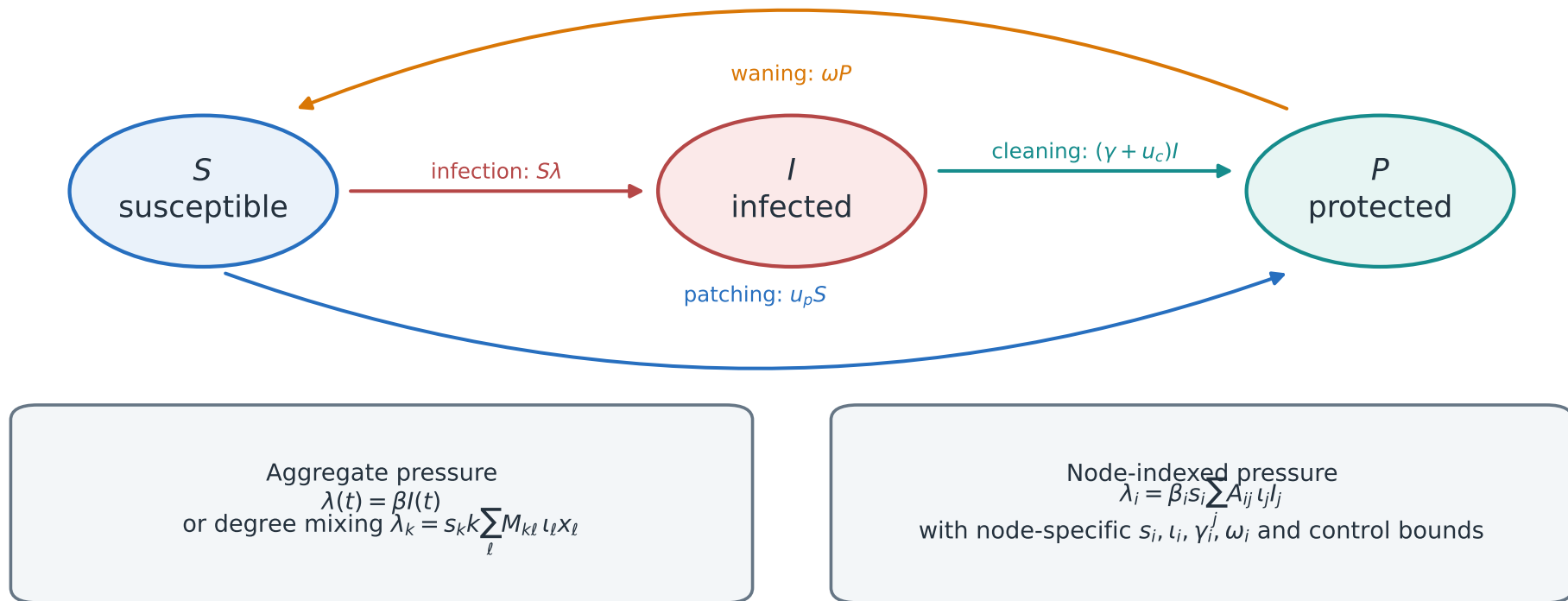


Canonical SIPS state transitions

State order is $[S, I, P]$ and $S + I + P = 1$; every flow transfers mass between compartments.



The invariant is a modeling check, not an extra loss term in the ODE solver.